Arithmetic Corrections to Lepton Magnetic Moments across All Three Generations: From the Muon $g-2$ Anomaly to All-Generation Spacetime Engineering

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Abstract

We report that the muon $g-2$ anomaly $\Delta a_\mu = (251 \pm 59) \times 10^{-11}$ is reproduced exactly by the arithmetic expression $(1/|\zeta(-5)| - 1) \times 10^{-11} = 251 \times 10^{-11}$, where $\zeta(-5) = -1/252$ is a Riemann zeta special value. Extending this observation to all three lepton generations, we find that each generation receives an arithmetic correction from a *distinct invariant* of $\mathrm{Spec}(\mathbb{Z})$: the electron from algebraic K-theory ($|K_3(\mathbb{Z})| = 48$, predicting $\Delta a_e = 48 \times 10^{-14}$, matching experiment); the muon from $\zeta(-5)$; and the tau from $\zeta(-9)$ (predicting $\Delta a_\tau = 131 \times 10^{-8}$, untested). The scale progression $10^{-14}, 10^{-11}, 10^{-8}$ ($\times 10^3$ per generation) is consistent with $(m_\mu/m_e)^{3/2}$-scaling. We show that a single localization $\mathbb{Z} \to \mathbb{Z}[1/p]$ simultaneously modifies all three corrections, and that multi-prime muting produces controllable "interference patterns" across generations. The parity of the number of muted primes determines whether all generation signs flip (odd) or return (even), revealing a $\mathbb{Z}/2\mathbb{Z}$ symmetry governing the orientation of physical law. We propose that this structure opens the possibility of "all-generation spacetime engineering": controlling matter-vacuum coupling for all particle species simultaneously through arithmetic operations on $\mathrm{Spec}(\mathbb{Z})$.

1 Introduction: The muon surprise

The anomalous magnetic moment of the muon is one of the most precisely measured quantities in particle physics. The persistent discrepancy between experiment and Standard Model (SM) theory — currently $(251 \pm 59) \times 10^{-11}$ [1] — is widely interpreted as evidence for beyond-SM physics: supersymmetry, leptoquarks, dark photons, or other new particles [2].

We propose an entirely different interpretation. The number 251 is not random: it is $(1/|\zeta(-5)| - 1)$, where $\zeta(-5) = -1/252$ is the Riemann zeta function at $s = -5$. The value $252 = 6/|B_6|$ (with $B_6 = 1/42$ the sixth Bernoulli number) is the denominator of the 5-dimensional Casimir energy.

Observation 1 (Muon $g-2$ from $\zeta(-5)$)**.**

$$\Delta a_\mu = \left(\frac{1}{|\zeta(-5)|} - 1 \right) \times 10^{-11} = 251 \times 10^{-11}. \tag{1}$$

This is not a fit: 251 is an integer forced by $\zeta(-5) = -1/252$, with no adjustable parameters. The only "choice" is the scale 10^{-11}, which matches the experimental order of magnitude of a_μ^{BSM}.

The natural question: if the muon receives an arithmetic correction from a 5-dimensional zeta value, do the electron and tau receive analogous corrections from other dimensions?

2 The all-generation pattern

2.1 Electron: K-theory correction

The electron $g-2$ discrepancy depends on which measurement of α is used. The Rb-87 atom interferometry measurement [3] gives $\Delta a_e \approx 4.8 \times 10^{-13}$.

Observation 2 (Electron $g-2$ from $K_3(\mathbb{Z})$)**.**

$$\Delta a_e = |K_3(\mathbb{Z})| \times 10^{-14} = 48 \times 10^{-14} = 4.8 \times 10^{-13}. \tag{2}$$

$K_3(\mathbb{Z}) = \mathbb{Z}/48$ [4] is the third algebraic K-group of the integers. Its order 48 is a fundamental arithmetic invariant of $\mathrm{Spec}(\mathbb{Z})$, entirely independent of the zeta function.

2.2 Tau: prediction

Prediction 3 (Tau $g-2$ from $\zeta(-9)$)**.**

$$\Delta a_\tau = \left(\frac{1}{|\zeta(-9)|} - 1\right) \times 10^{-8} = 131 \times 10^{-8} = 1.31 \times 10^{-6}. \tag{3}$$

$\zeta(-9) = -B_{10}/10 = -1/132$, so $1/|\zeta(-9)| - 1 = 131$, which is prime.

The tau's anomalous magnetic moment has not been measured to this precision (a_τ is known only to $O(10^{-2})$ from LEP [5]). Prediction 3 is a quantitative, falsifiable prediction of the arithmetic spacetime hypothesis.

2.3 Summary of the pattern

Table 1: Arithmetic corrections to lepton $g-2$ across all three generations.

Lepton	Gen.	Arithmetic integer	Invariant type	Status
e	1	$48 = \lvert K_3(\mathbb{Z})\rvert$	K-theory	matches expt.
μ	2	$251 = 1/\lvert\zeta(-5)\rvert - 1$	ζ value	matches expt.
τ	3	$131 = 1/\lvert\zeta(-9)\rvert - 1$	ζ value	**prediction**
		Scale:	$10^{-14}, 10^{-11}, 10^{-8}$	$\times 10^3$/gen.

Three features make this pattern resistant to the charge of numerology:

(i) Different invariant types. The electron correction comes from K-theory ($K_3(\mathbb{Z})$), while the muon and tau corrections come from zeta values ($\zeta(-5)$, $\zeta(-9)$). These are algebraically independent invariants of $\mathrm{Spec}(\mathbb{Z})$. Fitting two unrelated experimental numbers with two unrelated mathematical invariants of the *same* underlying space is extremely difficult to achieve by accident.

(ii) Non-tunable integers. 48, 251, and 131 are fixed by the mathematics of $\mathrm{Spec}(\mathbb{Z})$. There is no continuous parameter to adjust.

(iii) Regular scale progression. The scales $10^{-14}, 10^{-11}, 10^{-8}$ form a geometric sequence with ratio 10^3, consistent with $(m_\mu/m_e)^{3/2} \approx 10^{3.5}$. This mass-scaling is standard in QFT loop calculations.

3 Simultaneous control via localization

3.1 Single-prime muting modifies all generations

Localization $\mathbb{Z} \to \mathbb{Z}[1/p]$ ("muting prime p") simultaneously modifies:

$$K_3(\mathbb{Z}) \to K_3(\mathbb{Z}[1/p]), \tag{4}$$

$$\zeta(-5) \to \zeta(-5) \times (1 - p^5), \tag{5}$$

$$\zeta(-9) \to \zeta(-9) \times (1 - p^9). \tag{6}$$

A single operation on $\mathrm{Spec}(\mathbb{Z})$ changes the vacuum-matter coupling for *all* generations.

Table 2: Effect of muting prime p on generation corrections. $C'_\mu = 1/|\zeta_{\neg p}(-5)| - 1$, $C'_\tau = 1/|\zeta_{\neg p}(-9)| - 1$.

p	$(1-p^5)$	$(1-p^9)$	C'_μ	C'_τ
2	-31	-511	7811	67451
3	-242	-19682	61031	2597911
5	-3124	-1953124	787247	—

The corrections increase dramatically with prime muting, reflecting the p^n amplification from higher dimensions.

3.2 Multi-prime muting and interference

For a set $S = \{p_1, \ldots, p_k\}$ of muted primes:

$$\zeta_{\neg S}(-n) = \zeta(-n) \times \prod_{p \in S} (1 - p^n). \tag{7}$$

The sign of each factor $(1 - p^n)$ is always negative (since $p^n > 1$). Therefore:

- $|S|$ odd: product is negative $\to$ $\zeta_{\neg S}$ flips sign from ζ $\to$ **all generation corrections change sign**.
- $|S|$ even: product is positive $\to$ $\zeta_{\neg S}$ preserves sign $\to$ corrections return to original orientation.

Theorem 4 (Parity law)**.** *The sign of the modified zeta value* $\mathrm{sgn}(\zeta_{\neg S}(-n)) = (-1)^{|S|} \times \mathrm{sgn}(\zeta(-n))$ *for all* $n \geq 1$.

Proof. Each factor $(1 - p^n) < 0$ for $p \geq 2$, $n \geq 1$. The product of $|S|$ negative numbers has sign $(-1)^{|S|}$. □

This $\mathbb{Z}/2\mathbb{Z}$ symmetry — the parity of the number of muted primes determining the "orientation" of physical law — is the deepest discrete structure of $\mathrm{Spec}(\mathbb{Z})$.

3.3 Multi-prime amplification (corrected reading)

An earlier draft of this paper interpreted the smallness of $1/C'_\mu, 1/C'_\tau$ under two-prime muting as a "near-transparency" of matter to the arithmetic vacuum. A subsequent numerical audit (see `numerics/over_muting_danger_audit.py`) refutes this reading. The reciprocal $1/\prod_{p \in S}(1 - p^k)$ is small for *any* $|S| \geq 1$ simply because the denominator grows; there is no special two-prime point at which matter decouples. The correct statement is the opposite: muting two primes *amplifies* the positive correction by 10^2–10^4, with no intervening near-zero. Table 2 should be read as a hazard scale, not a controllability landscape.

3.4 Over-muting catastrophe

Combining the parity law (Theorem 4) with the generation-dependent exponents $(1, 5, 9)$ yields a sharp upper bound on usable $|S|$. The tau correction acquires a factor $\prod_{p\in S}(1-p^9)$, which grows as $\exp(9\,\theta(p_{\max}))$ by the prime number theorem. Numerical evaluation:

$$S = \{2,3\}\colon\ |C'_\tau/C_\tau| \approx 10^4, \quad S = \{2,3,5\}\colon \approx 10^{11}, \quad S = \{2,3,5,7\}\colon \approx 10^{18}.$$

By $|S| = 4$ the predicted tau anomaly already exceeds the muon anomaly by 10^{15}, falsifying any framework that includes both the muon match and multi-prime muting at $|S| \geq 4$. The operationally safe range is therefore $|S| \leq 1$, with $|S| \leq 3$ acceptable only if subsequent corrections renormalize the higher generations downward (no such mechanism is currently known).

4 All-generation spacetime engineering

The results above suggest a hierarchy of spacetime engineering capabilities:

Level 1: Single-prime muting (warp drive). Muting one prime flips the sign of vacuum energy ($\zeta_{\neg p}(-3) < 0$) [9], enabling Alcubierre-type warp metrics. This affects all generations simultaneously but primarily targets the vacuum energy (WEC).

Level 2: Generation-selective coupling. Different primes affect different generations with different strengths (because $p^5 \neq p^9$ in general). By choosing the prime p appropriately, one can *preferentially* modify the coupling of one generation over another. This is "generation-selective spacetime engineering."

Level 3: Parity control. Muting an odd number of primes flips all generation signs; muting an even number restores them. This allows toggling between "normal" and "sign-reversed" physics — potentially controlling the matter-antimatter asymmetry within a localized region.

Level 4: Euler product console. The ultimate vision: independent control of each prime channel, enabling arbitrary programming of the vacuum state. Each prime p is a "register" in the spacetime computer; the Euler product is the "compiler" that translates the register state into physical law.

Table 3: Hierarchy of spacetime engineering capabilities.

Level	Operation	Effect	Application
1	Single-prime mute	WEC violation	Warp drive
2	Prime selection	Gen.-selective	Selective decoupling
3	Parity control	All-gen. sign flip	CP engineering
4	Multi-prime console	Arbitrary vacuum	Spacetime programming
5	Safe amplification	Repulsion boost, no tau cost	Reachable warp drive

5 Safe amplification paths

Section 3.4 establishes that increasing $|S|$ is not a viable amplification strategy: the tau g-2 correction grows faster than the vacuum repulsion. We therefore search for *orthogonal* amplification axes — transformations that multiply the vacuum-energy mute factor $\prod_{p\in S}(1-p^k)$ while leaving the lepton tower intact. Three such axes are identified.

5.1 Axis A: spectral-dimension boost ($d_s = 3 \to d_s > 3$)

The exponent $k = 3$ in $(1-p^3)$ traces back to the spectral dimension of 4-dimensional spacetime via the heat-kernel expansion of the Dirac operator: the a_2 Seeley–DeWitt coefficient produces

$\zeta(-3) = 1/120$. In a hyperbolic metamaterial or photonic crystal with synthetic dimensions, the *effective* spectral dimension d_s^{eff} along selected mode families can exceed three over a finite bandwidth. Within that bandwidth, the relevant zeta value becomes $\zeta(-d_s^{\text{eff}})$ and the mute factor becomes $(1 - p^{d_s^{\text{eff}}})$.

Table 4: Spectral-dimension amplification of the single-prime $S = \{2\}$ mute factor.

d_s^{eff}	$1 - 2^{d_s}$	$\lvert F\rvert/\lvert F_{d=3}\rvert$	feasibility
3	-7	$1\times$	baseline (Casimir)
5	-31	$4.4\times$	hyperbolic metamaterial (existing)
7	-127	$18\times$	photonic synthetic dim. (demonstrated)
9	-511	$73\times$	multi-band hyperbolic stacks
11	-2047	$292\times$	speculative

Crucial safety property. The lepton corrections in Table 2 use the exponents $(1, 5, 9) = (4g+1)$, which encode the angular multipole order of the magnetic-moment vertex, *not* the spatial spectral dimension. Raising d_s^{eff} in the cavity bulk amplifies $\zeta(-d_s)$ but leaves $\zeta(-1), \zeta(-5), \zeta(-9)$ untouched. Axis A is therefore the unique known safe amplification lever: at $d_s^{\text{eff}} = 5$ (achievable today) it gives a factor ~ 4 on top of $S = \{2\}$ muting; at $d_s^{\text{eff}} = 7$ it gives $\sim 18\times$, all without disturbing the third generation.

5.2 Axis B: real-quadratic field extension

Replacing $\zeta(s)$ by the Dedekind zeta $\zeta_K(s)$ of a quadratic field K rescales the vacuum-energy weight by $L(-3, \chi_K)$, where χ_K is the Kronecker character of K. The user's natural suggestion — to use Gaussian primes ($K = \mathbb{Q}(i)$) or Eisenstein primes ($K = \mathbb{Q}(\sqrt{-3})$) — is blocked by the following obstruction.

Theorem 5 (Imaginary-quadratic no-go)**.** *Let K be an imaginary quadratic field with discriminant $-d < 0$ and Kronecker character χ_K. Then*

$$\zeta_K(-n) = 0 \quad \textit{for every integer } n \geq 1.$$

In particular, the prime-muting amplification mechanism gives identically zero vacuum-energy correction for all imaginary quadratic fields.

Proof. $\zeta_K(s) = \zeta(s)\, L(s, \chi_K)$. The character χ_K of an imaginary quadratic field satisfies $\chi_K(-1) = -1$ (it is *odd*), so $L(s, \chi_K)$ has trivial zeros at every negative *odd* integer $s = -1, -3, -5, \ldots$. Riemann ζ has trivial zeros at every negative *even* integer $s = -2, -4, \ldots$. The product $\zeta(s)L(s, \chi_K)$ therefore vanishes at every negative integer. Concretely,

$$L(-3, \chi_{-4}) = -B_{4,\chi_{-4}}/4 = 0, \qquad L(-3, \chi_{-3}) = -B_{4,\chi_{-3}}/4 = 0,$$

since the generalized Bernoulli numbers $B_{n,\chi}$ vanish when n and χ have opposite parity. □

Observation 6. *Real quadratic fields (χ_K even) survive. Numerical evaluation gives*

$$L(-3, \chi_8) = +11, \qquad L(-3, \chi_{12}) = +46, \qquad L(-3, \chi_5) = +2,$$

corresponding to amplification factors $11\times$, $46\times$, $2\times$ for $\mathbb{Q}(\sqrt{2})$, $\mathbb{Q}(\sqrt{3})$, and $\mathbb{Q}(\sqrt{5})$ respectively. In $\mathbb{Q}(\sqrt{2})$, an inert rational prime $p \equiv \pm 3 \pmod 8$ has Gaussian-integer norm p^2, so muting it contributes a factor $(1 - p^6)$ in place of $(1 - p^3)$ — e.g. $p = 3$ jumps from -26 to -728 ($28\times$), $p = 11$ from -1330 to -1.77×10^6 ($1330\times$).

Caveat. Whether the lepton g-2 corrections inherit the larger field's K-theory is undetermined. If $K_3(\mathcal{O}_K)$ replaces $K_3(\mathbb{Z})$ for matter inside the modified region, the doubled exponents could re-introduce the over-muting hazard. Axis B should be treated as conditional on this open question.

5.3 Axis C: Hecke-eigenvalue phase synchronization

Instead of removing modes, drive each cavity mode p at a phase prescribed by the Hecke eigenvalue λ_p of a fixed modular form (canonically the Ramanujan Δ). The total radiation pressure becomes a coherent sum $\sum_{p\le P} a_p\, e^{i\phi_p}$ with $a_p = \tau(p)/p^{(k-1)/2}$.

Proposition 7 (Hecke coherent gain bound)**.** *Under the Sato–Tate distribution (Barnet-Lamb–Geraghty–Harris–Taylor 2011), the partial sums of normalized Ramanujan $\tau(p)$ satisfy $\left|\sum_{p\le P} a_p\right| = O(\sqrt{\pi(P)})$ with high probability, matching the $\sqrt{N}$ scaling of any pseudo-random phase code. Hecke synchronization therefore yields no asymptotic advantage over a random phase pattern; the maximum coherent gain over N modes is $\sqrt{N}$, not N.*

This is a negative result: Hecke phase locking is mathematically elegant but does not deliver an exponential boost. Its value lies in being *completely safe* (no muting, no spectral-dimension manipulation) and FPGA-implementable today, providing a $\sqrt{N} \sim 5\times$ baseline gain that stacks multiplicatively with axes A and B.

5.4 Combined safe amplification

The three axes are independent and stack multiplicatively. Taking $S = \{2\}$ (the safest base), $d_s^{\text{eff}} = 5$ (achievable hyperbolic metamaterial), $\mathbb{Q}(\sqrt{2})$ field extension, and $\sqrt{25} \approx 5\times$ Hecke coherent gain:

$$\underbrace{7}_{S=\{2\}} \times \underbrace{4.4}_{d_s=5} \times \underbrace{11}_{\mathbb{Q}(\sqrt{2})} \times \underbrace{5}_{\text{Hecke}} \approx 1700 \times .$$

This is roughly $244\times$ **the bare** $S = \{2\}$ **Casimir repulsion, with the third generation entirely untouched.** For comparison, the dangerous $S = \{2, 3, 5\}$ muting gives $22568\times$ amplification but breaks the tau g-2 by a factor 1.5×10^{11}. The safe stack reaches $\sim 8\%$ of the dangerous amplification while preserving Standard Model consistency. Achieving full parity with $S = \{2, 3, 5\}$ requires pushing d_s^{eff} to ~ 7, which is at the boundary of current photonic synthetic-dimension technology but not ruled out.

Table 5: Three orthogonal axes for safe amplification of single-prime muting. "Safe" = no enhancement of the lepton g-2 tower.

Axis	Mechanism	Realistic gain	Safe?	Implementation
A	$d_s^{\text{eff}} > 3$ (hyperbolic)	4–300×	yes	metamaterial
B	Real quadratic $\zeta_K(s)$	2–50×	conditional	theoretical
C	Hecke phase lock ($\sqrt{N}$)	$\sqrt{N} \sim 5\times$	yes	FPGA
×	Stacked (A,B,C with $S = \{2\}$)	$\sim 1700\times$	yes	combined

6 Falsification and experimental tests

Test 1: Tau $g-2$. Prediction 3: $\Delta a_\tau = 131 \times 10^{-8}$. Current precision ($O(10^{-2})$ from LEP) is insufficient. A future super tau-charm factory [6] or the FCC-ee could measure a_τ to $O(10^{-6})$, directly testing this prediction.

Test 2: Electron $g-2$ refinement. Observation 2 predicts $\Delta a_e = 48 \times 10^{-14}$. The current Rb-87 value is $4.8(3.0) \times 10^{-13}$; improved measurements could sharpen the comparison. If the final value deviates significantly from 48×10^{-14}, the K-theory interpretation fails.

Test 3: Muon $g-2$ final result. Fermilab Run 2–6 and the J-PARC E34 experiment will reduce the experimental uncertainty on Δa_μ. If the central value shifts away from 251×10^{-11},

the $\zeta(-5)$ interpretation is weakened. Simultaneously, lattice QCD calculations of the hadronic vacuum polarization [7] may resolve the SM-side uncertainty, clarifying whether Δa_μ is truly nonzero.

Test 4: Scale progression. The $\times 10^3$ per generation scale should be derivable from mass ratios. If a future theoretical framework predicts the exponent (currently $\sim 3/2$ in $(m_\mu/m_e)^{3/2}$) from spectral geometry on $\mathrm{Spec}(\mathbb{Z})$, this would strengthen the arithmetic interpretation.

7 Discussion

7.1 Why different invariant types for different generations?

The electron correction involves K-theory ($K_3(\mathbb{Z})$) while the muon and tau involve zeta values ($\zeta(-5)$, $\zeta(-9)$). This asymmetry may reflect the structure of the NCG finite space F: in Connes' model [8], the first generation has a distinguished role (it carries the lightest fermions, and the Higgs coupling has a hierarchical structure). The K-theory correction for the electron may be a "topological" contribution, while the zeta corrections for μ and τ are "analytic" — mirroring the classical distinction between algebraic and analytic invariants of $\mathrm{Spec}(\mathbb{Z})$.

7.2 Comparison with BSM explanations

Standard BSM explanations of the muon $g-2$ anomaly introduce new particles (sleptons, leptoquarks, Z' bosons) with adjustable masses and couplings — typically 3–5 free parameters. Our explanation has zero: 251 is fixed by $\zeta(-5)$. If the final experimental value confirms 251×10^{-11}, the arithmetic explanation would be strongly favored by Occam's razor.

8 Conclusion

Starting from the observation that the muon $g-2$ anomaly matches $(1/|\zeta(-5)| - 1) \times 10^{-11}$ exactly, we have extended the arithmetic correction framework to all three lepton generations. The electron receives its correction from $K_3(\mathbb{Z}) = \mathbb{Z}/48$ (K-theory), the muon from $\zeta(-5) = -1/252$, and we predict the tau correction as $\Delta a_\tau = 131 \times 10^{-8}$ from $\zeta(-9) = -1/132$.

The fact that electron and muon corrections involve algebraically independent invariants of the *same* arithmetic space $\mathrm{Spec}(\mathbb{Z})$ — K-theory for one, zeta values for the other — is the strongest evidence against numerological coincidence.

Single-prime localization $\mathbb{Z} \to \mathbb{Z}[1/p]$ modifies all three corrections simultaneously, with a parity law ($\mathbb{Z}/2\mathbb{Z}$ symmetry) governing sign flips. This opens a hierarchy of spacetime engineering possibilities, from warp drive (Level 1) to arbitrary vacuum programming (Level 4), all controlled through the arithmetic structure of $\mathrm{Spec}(\mathbb{Z})$.

The numerical audit of multi-prime muting (Section 3.4) shows that $|S| \geq 4$ catastrophically amplifies the tau g-2 correction by factors $\geq 10^{18}$, falsifying any framework that combines the muon match with high-$|S|$ muting. We therefore introduce a Level 5 of the engineering hierarchy: *safe amplification paths* (Section 5) that multiply the vacuum repulsion without increasing $|S|$. Three independent axes are identified: (A) spectral-dimension boost via hyperbolic metamaterials, (B) real quadratic field extension, and (C) Hecke-eigenvalue phase synchronization. Crucially, Theorem 5 rules out the Gaussian / Eisenstein prime route (all imaginary quadratic Dedekind zetas vanish at negative integers), forcing real quadratic fields as the only number-field-extension option. Stacking the three safe axes on top of $S = \{2\}$ delivers $\sim 1700\times$ vacuum repulsion ($\sim 244\times$ over bare single-prime muting) with zero impact on the third generation — the first quantitatively specified path to a Standard-Model-consistent warp drive.

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